

Path to A — DATA 422 Final Capstone Plan

Automated CTF Program (Cadet-Initiated)

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0. Document Purpose

This is your day-by-day execution plan from **today (Wed 29 Apr 2026)** through the **Final Paper deadline (COB Fri 8 May 2026)**. Every step is tagged with the rubric criterion it earns and the artifact it produces. Code blocks are ready to paste — modify paths only.

The plan assumes:

- You brief on the **earliest** Final Outbrief slot (Mon 4 May, T38). If you brief later (T39 = Wed 6 May or T40 = Fri 8 May) the buffer days at the end grow.
- You will use a **hybrid backend**: OpenAI API for the bulk benchmark sweep, Ollama (gemma4:26b) for a smaller subset that anchors the local-model thread.
- This is a **cadet-initiated, individual capstone** with Lt Col Todd as mentor (no external sponsor). The “decision-quality recommendations” rubric line maps to recommendations for (a) Lt Col Todd’s research interests, (b) future cadet researchers continuing this work, and (c) an explicit “what would a master’s thesis follow-on look like” section.

1. How a Cadet-Initiated Capstone Changes the Rubric Mapping

The Course Letter and the rubrics use the word “sponsor” frequently. For your project there is no operational sponsor — Lt Col Todd is your mentor. Three mappings to make explicit so the panel doesn’t penalize for missing what isn’t there:

Rubric line	What “sponsor” maps to in your case
“Decision-quality information that directly solves sponsor needs”	(a) Recommendations <i>to Lt Col Todd</i> on follow-on research directions; (b) deployment envelope for a hypothetical

Rubric line	What “sponsor” maps to in your case
	operational user; (c) explicit master’s thesis problem statements derived from your gaps
“Open, transparent, timely communication with sponsor”	Documented mentor meetings (cite dates in the paper acknowledgements)
“Sponsor approval” for cloud APIs (Course Letter §9)	Mentor approval — you still owe Lt Col Todd a heads-up on the OpenAI-API decision before you spend
“Has answered the questions your sponsor was asking”	Has answered your own three research questions; restate them verbatim in §6 of the paper and on the conclusions slide

In the Final Brief and Paper, **state once, near the top, that this is a cadet-initiated individual capstone** so the panel scores recommendations against the right reference. Suggested wording: *“This capstone is a cadet-initiated individual research project mentored by Lt Col Todd (DFCS). Research recommendations are scoped to follow-on work and future cadet researchers in this domain rather than an external operational sponsor.”*

2. Strategic Reframe (Read This First)

Your mid-semester brief asked: “*Can a 26B local model + lessons-learned KB match capability?*” and reported the honest negative result that KB hurt overall solve rate (3 regressed, 2 improved, 9 efficient).

For an **A**, that finding alone is insufficient — the rubric explicitly penalizes narratives that dwell on the journey/failures over the final solution. **Reframe the project around three claims that the data can support:**

Claim 1 — Scaffolding-tier hypothesis. Scaffolding investment scales inversely with model capability. At 26B local, scaffolding moves the needle materially. At frontier (GPT-4o), scaffolding marginal value approaches zero. Lessons-DB is one component of scaffolding and behaves accordingly.

Claim 2 — Regression diagnosis. Where lessons-DB hurt, the cause was mechanical (context bloat, stale rule cross-talk, narrow IDOR enumeration default). v3.10 P0–P5 patches address each mechanism; post-fix benchmark quantifies the remediation.

Claim 3 — Capability envelope. Across N challenges in M categories, the solver reliably solves categories X/Y/Z; categories P/Q remain open. This becomes the deployment recommendation table.

These three claims map cleanly to the three research questions on slide 8 of your mid-semester deck and convert the “did RAG work?” framing into a “where does scaffolding pay off, and how do we diagnose its failures?” framing — which is exactly what the rubric A-line (“explored novel methodologies, integrated them, comprehensive discussion”) rewards.

3. Calendar — Day-by-Day

Day	Date	Focus	Hrs	Deliverable
Wed	29 Apr	Bench setup; mentor email; lit review pull	6	bench/challenges.tsv, scripts/benchmark.sh, batch 1 kicked off
Thu	30 Apr	Lit review synthesis; methods inventory; bench batch 2	4+idle	refs/synthesis.md, methods_inventory.md, partial CSV
Fri	1 May	Mentor sync; figures; regression diagnosis	6	out/figures/ fig{1..5}.png, results/ regression_diagnoses.md, sponsor_notes.md
Sat	2 May	Paper draft §1–4 (Abstract, Intro, Lit, Methods)	6	final_paper/main.tex partial
Sun	3 May	Paper draft §5–6 (Results, Conclusions); citations	6	final_paper/main.tex complete first draft, references.bib
Mon	4 May	Slide deck (21 slides)	5	presentation/ final_brief.pptx
Tue	5 May	Q&A prep; solo run- through	4	qa_prep.md, timed solo
Wed	6 May	Mock run- through; README handoff; paper edit pass 1	4	README_HANDOFF.md, paper rev 2
Thu	7 May	Paper edit pass 2; AI- usage statement;	3	paper rev 3, slides final

Day	Date	Focus	Hrs	Deliverable
		final slide polish		
Fri	8 May	Submit by 0900	1	Email to Maj Merrick

Total: ~45 hours. Tight but doable. Lighter days near the brief give buffer.

4. Phase 1 — Generate Evidence (Wed 29 Apr → Fri 1 May)

4.1 Build the benchmark spec (Wed AM, 1 hr)

Create `bench/challenges.tsv` (tab-separated, columns: name url category preset notes):

```
# Name<TAB>URL<TAB>category<TAB>preset<TAB>notes
crystal-peak<TAB>http://crystal-
peak.picoctf.net:<port><TAB>idor<TAB>picoctf<TAB>md5(uid) IDOR
# ... continue with the 18 mid-sem challenges + 5-7 fresh picoCTF
retired
```

Sourcing the 18: `ls out/lessons_knowledge/lessons_*.md | sed 's/.*lessons_[0-9]*_//' | sed 's/_r[0-9]*_.*//' | sort -u` — each file name is a challenge slug.

Adding new ones: pick from picoCTF retired challenges (<https://play.picoctf.org/practice>) plus any leftover from your `batch_*` directories. Stratify so each category has ≥ 2 entries.

Document scope/exclusions in `bench/README.md` — the rubric A-line wants “data source, scope, and limitations clearly described.”

4.2 The benchmark harness (Wed AM, 1.5 hr)

Three new files. All three are designed to be paste-ready.

4.2.1 `scripts/benchmark.sh`

```
#!/usr/bin/env bash
# Hybrid CTF benchmark – runs every challenge under three rag-mode
# configs
# on each backend tier. Tier 1: OpenAI (frontier). Tier 2: Ollama
# (26B local).
# Outputs runs/benchmark_<ts>/<tier>/<config>/<name>.log +
# metrics.csv.
set -euo pipefail
```

```

HERE="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
TS=$(date +%Y%m%d_%H%M%S)
BENCH_ROOT="${HERE}/runs/benchmark_${TS}"
TSV="${HERE}/bench/challenges.tsv"
CONFIGS=("none" "lessons_readonly" "lessons_write")

# --- Tier toggles
-----
RUN_OPENAI="${RUN_OPENAI:-1}"          # set to 0 to skip
RUN_OLLAMA="${RUN_OLLAMA:-1}"          # set to 0 to skip
OPENAI_MODEL="${OPENAI_MODEL:-gpt-4o-mini}" # cheap default
OLLAMA_MODEL="${OLLAMA_MODEL:-gemma4:26b}"
MAX_STEPS="${MAX_STEPS:-15}"
# Limit Ollama to a 6-challenge subset (rotate via OLLAMA_SUBSET
# env).
OLLAMA_SUBSET="${OLLAMA_SUBSET:-crystal-peak,verbal-stack,login-
token,jwt-mate,cookie-monster,ssti-warmup}"

run_one() {
    local tier="$1"; local provider="$2"; local model="$3"; local
    cfg="$4"
    local name="$5"; local url="$6"; local preset="$7"
    local out="${BENCH_ROOT}/${tier}/${cfg}/${name}.log"
    mkdir -p "$(dirname "$out")"
    "${HERE}/scripts/run.sh" \
        --challenge-url "$url" \
        --challenge-name "$name" \
        --description "Benchmark: $name" \
        --flag-preset "$preset" \
        --llm-provider "$provider" \
        --model "$model" \
        --max-steps "$MAX_STEPS" \
        --rag-mode "$cfg" \
        --history-window-size 16 \
        > "$out" 2>&1 || echo "[WARN] $tier/$cfg/$name exit non-
zero"
}

while IFS=$'\t' read -r name url category preset notes; do
    [[ "$name" =~ ^# ]] && continue
    [[ -z "$name" ]] && continue

    if [[ "$RUN_OPENAI" == "1" ]]; then
        for cfg in "${CONFIGS[@]"; do
            run_one "openai" "openai" "$OPENAI_MODEL" "$cfg"
            "$name" "$url" "$preset"
        done
    fi

    if [[ "$RUN_OLLAMA" == "1" && ",${OLLAMA_SUBSET}," == *,${
{name},* ]]; then
        for cfg in "${CONFIGS[@]"; do
            run_one "ollama" "ollama" "$OLLAMA_MODEL" "$cfg"
            "$name" "$url" "$preset"

```

```

        done
    fi
done < "$TSV"

# Extract metrics
"${HERE}/.venv/bin/python" "${HERE}/scripts/extract_metrics.py"
    "$BENCH_ROOT" \
    > "${BENCH_ROOT}/metrics.csv"
echo "Benchmark complete. Results: ${BENCH_ROOT}/"

chmod +x scripts/benchmark.sh.

```

4.2.2 scripts/extract_metrics.py

```

"""Walk a benchmark output tree and emit a CSV with one row per
(tier, config, challenge). Captures solve outcome, steps,
    validity,
repeated-call rate, and wall-clock estimate from each run.log."""
from __future__ import annotations
import csv
import re
import sys
from pathlib import Path

FLAG_RE = re.compile(r"\[FLAG DETECTED\] (.+)")
STEP_RE = re.compile(r"^--- Step (\d+)/\d+ ---")
ACTION_RE = re.compile(r"^Action: Using tool '([^']+)'")
ERROR_OBS_RE = re.compile(r"^Observation:.*result=error")
PARSE_FAIL_RE = re.compile(r"Could not parse simplified ReAct
    response")

def parse_one(path: Path) -> dict:
    text = path.read_text(errors="replace")
    flag = FLAG_RE.search(text)
    steps = [int(m.group(1)) for m in STEP_RE.finditer(text)]
    actions = ACTION_RE.findall(text)
    error_obs = len(ERROR_OBS_RE.findall(text))
    parse_fails = len(PARSE_FAIL_RE.findall(text))
    repeated = 0
    if actions:
        seen = {}
        for a in actions:
            seen[a] = seen.get(a, 0) + 1
        repeated = sum(c - 1 for c in seen.values() if c > 1)
    return {
        "max_step_seen": max(steps) if steps else 0,
        "total_actions": len(actions),
        "unique_actions": len(set(actions)),
        "error_observations": error_obs,
        "parse_failures": parse_fails,
        "repeated_call_count": repeated,
        "outcome": "success" if flag else "failure",
    }

```

```

        "flag": flag.group(1) if flag else "",
    }

def main(root: str) -> None:
    rows = []
    for log in Path(root).rglob("*.log"):
        rel = log.relative_to(root)
        parts = rel.parts
        if len(parts) < 3:
            continue
        tier, cfg, name = parts[0], parts[1],
        parts[2].removesuffix(".log")
        m = parse_one(log)
        rows.append({"tier": tier, "config": cfg, "challenge":
            name, **m})
    cols = [
        "tier", "config", "challenge", "outcome", "max_step_seen",
        "total_actions", "unique_actions", "repeated_call_count",
        "error_observations", "parse_failures", "flag",
    ]
    w = csv.DictWriter(sys.stdout, fieldnames=cols)
    w.writeheader()
    w.writerows(sorted(rows, key=lambda r: (r["tier"],
        r["config"], r["challenge"])))

if __name__ == "__main__":
    main(sys.argv[1])

```

4.2.3 First dry-run on one challenge

```

echo -e "crystal-peak\thttp://crystal-
peak.picocf.net:<port>\tidor\tpicocf\ttest" > bench/
challenges.tsv
RUN_OLLAMA=0 OPENAI_MODEL=gpt-4o-mini MAX_STEPS=10 ./scripts/
benchmark.sh
cat runs/benchmark_*/metrics.csv | tail -5

```

If the CSV has a row with outcome=success you are good. Then expand the TSV to all 25 and run for real.

4.3 Run protocol

- **Wed PM, before bed:** kick off batch 1 with RUN_OLLAMA=0 (OpenAI only, ~25 challenges × 3 configs = 75 runs, ~2 hrs at \$5–10).
- **Thu AM:** check metrics.csv. If clean, kick off the Ollama subset (RUN_OPENAI=0 OLLAMA_SUBSET=...) — runs through Thu/Thu-night.
- **Fri AM:** combined CSV ready for analysis.

Cost guard. Set OPENAI_MODEL=gpt-4o-mini for the first pass. If you have budget left after results, re-run a subset on gpt-4o for an extra “frontier-tier” data point.

4.4 Mentor sync (Wed AM, 15 min email)

Send Lt Col Todd:

Subject: [CAPSTONE] 30-min vector check before final brief – Thu/Fri?

Sir,

Final brief is the week of 4 May. I've made significant changes to the

codebase since the 4 March update brief – most importantly:

- Patched the FAISS+OpenMP segfault that was preventing RAG runs
- Added 5 scaffolding fixes (v3.10) targeting failure modes I observed
 - on a recent gemma4:26b live run
- Adding a hybrid OpenAI-API arm to the benchmark to let me make cross-tier capability claims

I'd like 30 minutes this Thu or Fri to (a) walk you through the v3.10

results, (b) get your read on the reframed thesis, and (c) confirm the recommendations slide hits what you want to see.

Times that work for me: [list 3].

V/R,
Andrew

At the meeting, ask three questions:

1. *“For the recommendations slide, what’s the most useful thing this project could leave behind for follow-on cadet research?”*
2. *“Are you OK with me using OpenAI API for one arm of the benchmark? The data is all public picoCTF, but I want to flag it.”*
3. *“What’s the one thing you’d push back on if you saw this brief from another team?”*

Capture answers verbatim into sponsor_notes.md (keep the filename even though there’s no sponsor — your code already references it).

4.5 Lit-review additions (Thu, 3 hrs)

For each paper, write a 3-sentence synthesis paragraph in refs/synthesis.md:

1. **Shinn et al., “Reflexion: Language Agents with Verbal Reinforcement Learning,” NeurIPS 2023.** Justifies your compressed verbal lesson summaries.

2. **Zhao et al., “ExpeL: LLM Agents Are Experiential Learners,” AAAI 2024.** Justifies atomic rule docs vs. monolithic logs.
3. **Wang et al., “VOYAGER: An Open-Ended Embodied Agent with Large Language Models,” 2023.** Justifies the lessons-as-skills framing.
4. **Schick et al., “Toolformer: Language Models Can Teach Themselves to Use Tools,” NeurIPS 2023.** Anchors tool-use scaffolding broadly.
5. **Yang et al., “EnIGMA: Enhanced Interactive Generative Model Agent for CTF Challenges,” 2024.** Direct comparison point — what they did with GPT-4 vs. what your local-tier scaffolding does.
6. **Zhang et al., “Cybench: A Framework for Evaluating Cybersecurity Capabilities and Risks of Language Models,” 2024.** Frames your benchmark methodology.
7. **Wei et al., “Emergent Abilities of Large Language Models,” TMLR 2022.** Anchors your scaffolding-tier hypothesis (capability emerges at scale; sub-frontier models need help).
8. **Tann et al., “Using Large Language Models for Cybersecurity CTF Challenges,” 2023.** (Already in mid-sem — deepen the synthesis.)

Synthesis template per paper (3 sentences, no more):

[Authors, year, short title]. [One sentence: what they claim.] [One sentence: how this informs your design choice.] [One sentence: what your work does differently or adds.]

Example:

Zhao et al., 2024, ExpeL. They show that LLM agents extract higher-quality task heuristics when experiences are stored as atomic rule fragments rather than full trajectories, with retrieval gated by similarity to the current state. We adopt this atomic-rule structure in `failure_analyzer.py:generate_atomic_rule_doc` and add Jaccard-based deduplication on triggering conditions to prevent rule-set bloat across runs. Unlike ExpeL, our domain (web CTF) requires preserving exact attack inputs (e.g., template-injection payloads), so our extraction pipeline includes a post-scrub append of `winning_inputs` that bypasses the broad flag-regex redaction.

4.6 Novel-methods inventory (Wed PM, 30 min)

Create `methods_inventory.md` (this becomes one slide and one paper sub-section). For each item annotate which model tier it applies to:

Method	File	Local-tier	Frontier-tier
Hybrid BM25+vector retrieval (0.4/0.6, query-expand, rerank)	<code>rag/knowledge_base.py</code>	Y	Y
Outlines FSM grammar-constrained	<code>llm/adapters.py:MLXAdapter</code>	Y	n/a

Method	File	Local-tier	Frontier-tier
JSON (per-tool flat-enum)			
ExpeL-style atomic rule docs with confidence-ladder dedup	failure_analyzer.py	Y	Y
Reflexion compressed prior-lesson injection	runner.py:530–552	Y	Marginal
Jaccard site-fingerprint contamination filter	rag/knowledge_base.py	Y	Y
Function-calling via per-tool parameters_schema (Draft-07)	tools/schema.py	Y	Y
Tool family collapse (5 SQLi → 1 dispatcher)	tools/collapsed_*.py	Y	Marginal
Hash-pattern hint surface (v3.10 P1+P5a)	tools/core.py:_detect_hash_hints	Y	Marginal
StuckDetector with canonicalized JSON hashing (v3.10 P2)	tools/logging_wrapper.py	Y	Marginal
DYLD-preload Apple-Silicon FAISS+OpenMP fix	scripts/run.sh	Y	Y (env)

The “Local-tier vs. Frontier-tier” split is what makes Claim 1 (scaffolding scales inversely with capability) defensible — you’re showing exactly which methods stop mattering as the model gets stronger.

5. Phase 2 — Analysis & Writing (Sat 2 May → Sun 3 May)

5.1 Five figures (Fri PM, 3 hrs)

Save all to out/figures/ at 300 dpi. Caption each with the *finding*, not the contents.

```
# bench/analyze.py
import pandas as pd
import matplotlib.pyplot as plt
```

```

from pathlib import Path

df = pd.read_csv("runs/benchmark_LATEST/metrics.csv") # symlink
               or paste actual path
df["solved"] = (df["outcome"] == "success").astype(int)

# Fig 1 – Solve rate by tier × config × category
# (need to merge in category from bench/challenges.tsv)
ch = pd.read_csv("bench/challenges.tsv", sep="\t",
                 names=["challenge", "url", "category",
                        "preset", "notes"],
                 comment="#")
df = df.merge(ch[["challenge", "category"]], on="challenge",
             how="left")
fig, ax = plt.subplots(figsize=(10, 5))
pivot = (df.groupby(["category", "tier", "config"])["solved"]
        .mean().unstack(["tier", "config"]) * 100)
pivot.plot.bar(ax=ax)
ax.set_ylabel("Solve rate (%)")
ax.set_xlabel("Vulnerability category")
ax.set_title("Figure 1 – Scaffolding lift by category and model
             tier")
ax.legend(loc="upper right", fontsize=8)
plt.tight_layout()
plt.savefig("out/figures/fig1_solverate.png", dpi=300)

# Fig 2 – Per-challenge improvement matrix (heatmap)
import numpy as np
wide = df.pivot_table(index="challenge", columns=["tier",
          "config"], values="solved")
fig, ax = plt.subplots(figsize=(8, max(4, len(wide)*0.25)))
ax.imshow(wide.values, aspect="auto", cmap="RdYlGn", vmin=0,
          vmax=1)
ax.set_yticks(range(len(wide))); ax.set_yticklabels(wide.index,
          fontsize=8)
ax.set_xticks(range(len(wide.columns)));
          ax.set_xticklabels(["/".join(c) for c in wide.columns],
          rotation=45, ha="right", fontsize=8)
ax.set_title("Figure 2 – Per-challenge solve outcomes")
plt.tight_layout()
plt.savefig("out/figures/fig2_matrix.png", dpi=300)

# Fig 3 – Steps-to-flag CDF
solved = df[df["outcome"] == "success"]
fig, ax = plt.subplots(figsize=(7, 4))
for (tier, cfg), grp in solved.groupby(["tier", "config"]):
    s = sorted(grp["max_step_seen"])
    cdf = np.arange(1, len(s)+1) / len(s)
    ax.plot(s, cdf, label=f"{tier}/{cfg}", drawstyle="steps-post")
ax.set_xlabel("Steps to flag"); ax.set_ylabel("Cumulative solve
          fraction")
ax.set_title("Figure 3 – Efficiency: steps-to-flag CDF by config")
ax.legend(fontsize=8); plt.tight_layout()
plt.savefig("out/figures/fig3_cdf.png", dpi=300)

```

```

# Fig 4 – Tool-call validity rate (parse_failures vs.
    total_actions)
df["validity"] = 1 - (df["parse_failures"] /
    df["total_actions"].clip(lower=1))
fig, ax = plt.subplots(figsize=(8, 4))
df.groupby(["tier", "config"])
    ["validity"].mean().mul(100).plot.bar(ax=ax)
ax.set_ylabel("Tool-call validity (%)")
ax.set_title("Figure 4 – Tool-call validity rate by tier ×
    config")
plt.tight_layout()
plt.savefig("out/figures/fig4_validity.png", dpi=300)

# Fig 5 – Repeated-call rate (proxy for stuck loops)
fig, ax = plt.subplots(figsize=(8, 4))
df.groupby(["tier", "config"])
    ["repeated_call_count"].mean().plot.bar(ax=ax)
ax.set_ylabel("Mean repeated-call count per run")
ax.set_title("Figure 5 – Stuck-loop incidence by tier × config")
plt.tight_layout()
plt.savefig("out/figures/fig5_stuck.png", dpi=300)

```

Run with `.venv/bin/python bench/analyze.py`. Eyeball each figure — if any is degenerate (e.g. all categories tied), you have either a data problem or a finding worth investigating.

5.2 Causal regression analysis (Sat AM, 3 hrs)

For every challenge where lessons_write *worsened* outcome vs. baseline, write a results/regression_diagnoses.md entry:

```

### <challenge_slug>

**Observation.** Baseline (rag-mode=none) solved in N steps;
    lessons_write
solver did <not solve / solved in M>N steps>.

**Mechanism.** Diff of the trajectories at line ____ shows [stale
    lesson
cited / context bloat truncated key observation / off-task RAG
result
distracted from immediate signal].

**v3.10 fix that addresses this.** [P3a 1500-char proactive RAG
    cap / P5a
universal hash hint / P5b IDOR range_end=100 / etc.]

**Re-run result.** Post-fix benchmark on <date> shows
    <recovered / still
fails / improved by N steps>.

```

Aim for 3–5 entries (one per significant regression). This is **the difference between a B-grade “honest negative result” and an A-grade “diagnosed and partially remedied negative result.”**

5.3 Final paper (Sat PM → Sun, 6 hrs)

Use IEEE conference template (IEEEtran.cls, two-column). Skeleton:

```
final_paper/  
├── main.tex  
├── references.bib  
├── figures/    (symlink to ../out/figures)  
├── IEEEtran.cls  
└── IEEEtran.bst
```

Or use the Word IEEE template if LaTeX is too much to spin up. Either way, the **section content** below is what matters.

§1 Abstract (200 words, last thing you write)

Template: > CTF challenges are increasingly being used to evaluate the offensive-cyber > capability of LLM agents. This paper investigates the relative contribution > of *scaffolding* (prompt engineering, tool schemas, observation processing, > retrieval augmentation, and an experiential lessons-learned database) > versus raw model capability on web CTF tasks. We benchmark a unified agent > framework against N challenges across M vulnerability categories at two > capability tiers: gemma4:26b (local) and gpt-4o-mini (frontier-API). > Scaffolding lifts solve rate by X percentage points at the local tier and > Y percentage points at the frontier tier. We diagnose three mechanisms > (context bloat, stale-lesson contamination, narrow enumeration defaults) > by which a lessons-learned database can *worsen* solve rate, and we show > that targeted patches recover ~Z% of regressed cases. Code, benchmark, and > raw run logs are released for reproducibility. Our results suggest > scaffolding investment scales inversely with model capability — a finding > with practical implications for sub-frontier deployments.

Fill in the numbers from your CSV.

§2 Introduction (~1 page)

- Paragraph 1: Motivation (Volt Typhoon, NSA-MSS manpower asymmetry, AI-orchestrated cyber). Cite mid-sem references.
- Paragraph 2: Problem statement. State that this is a cadet-initiated individual capstone (per §1 of this plan).
- Paragraph 3: Three research questions, verbatim from mid-sem slide 8.
- Paragraph 4: Contributions, three bullets matching Claims 1/2/3 from §2 of this plan.
- Paragraph 5: Roadmap sentence.

§3 Theory & Literature Review (~1.5 pages)

Three threads (subsections), each ending in a *gap* sentence:

- §3.1 LLM CTF Agents — Tann 2023, EnIGMA 2024, Cybench 2024 → “no prior work evaluates a sub-frontier (26B) model with a structured experiential database on web-CTF tasks.”
- §3.2 Experiential Learning — Reflexion, ExpeL, VOYAGER → “prior work focuses on simulated environments and dense feedback; web CTF feedback is sparse and observation-rich, motivating the atomic-rule representation we use.”
- §3.3 Tool Use & Scaffolding at Tier — Toolformer, ReAct, Wei 2022 emergence → “the scaffolding-vs-capability tradeoff has not been empirically characterized at the 26B/frontier boundary; this paper does.”

§4 Methodology (~2 pages)

Subsections matching `methods_inventory.md`. Each subsection: 2–3 sentences of theory, one figure or code-snippet for the integration, one sentence of why that choice. **Cite the prior coursework** (e.g. DATA 350 / DATA 410 content if those covered RAG, vector search, agentic systems) — the rubric explicitly grades “integration of coursework tools with novel data science methodologies.”

§5 Results & Discussion (~2 pages)

Lead with Figure 1 + the headline number. Then Figures 2/3/4/5 in turn. End with the regression-diagnosis subsection. **Embrace the failed-model discussion** — the rubric A-line for the Update Brief was “fully embraces failed models and results”; the Final-Paper rubric implicitly inherits this via “comprehensive discussion of outputs and models.”

§6 Conclusions & Recommendations (~1 page)

Three subsections:

- §6.1 Answers to RQ1/RQ2/RQ3 — 2–3 sentences each, citing your figures.
- §6.2 Recommendations for Follow-on Research (your “sponsor” stand-in): five numbered items including a master’s-thesis problem statement derived from your gaps (e.g., “RL on top of SFT-LoRA over scaffold-aligned trajectories” from the audit).
- §6.3 Capability envelope — what this codebase reliably solves, deployment prerequisites, cost envelope.

§7 References

IEEE-numeric, every paper in `refs/synthesis.md` must appear, plus your mid-sem set.

§8 AI Usage Statement (Course Letter §9, mandatory)

Three-part: 1. **What:** Tools used (Claude Opus, GPT-4o, etc.) for code authorship, debugging, paper drafting. 2. **How:** Whether output was reviewed/edited/used verbatim. 3. **Where:** Link to representative transcript (your existing OneDrive transcript link works).

6. Phase 3 — Polish, Brief, and Ship (Mon 4 May → Fri 8 May)

6.1 21-slide deck structure (Mon, 5 hrs)

Use exactly this structure. ~70 sec/slide → 25 min total.

#	Slide	Rubric tie
1	Title (you, mentor, date, “cadet-initiated individual capstone”)	bookkeeping
2	Team (you + Lt Col Todd as mentor; thank Wilson, Goethal)	bookkeeping
3	Overview / agenda	bookkeeping
4	Motivation (Volt Typhoon, NSA-MSS gap, AI-cyber rise)	§1 Intro
5	Purpose statement + 3 research questions verbatim	§1 Intro
6	Lit Thread 1 — LLM CTF agents (gap statement bold)	§2 Lit
7	Lit Thread 2 — Experiential learning (gap)	§2 Lit
8	Lit Thread 3 — Tool use / capability tier (gap)	§2 Lit
9	Data — 25 challenges, stratification table, KB sources	§3 RQ&Data
10	Methodology — scaffolding architecture diagram	§4 Methods
11	Methodology — lessons-DB pipeline (atomic rules, Reflexion, fingerprint)	§4 Methods
12	Methodology — v3.10 fixes table (audit→implementation chain)	§4 Methods

#	Slide	Rubric tie
13	Headline result — Fig 1 + one-sentence finding	§5 Results
14	Fig 2 — per-challenge matrix	§5 Results
15	Fig 3 — efficiency CDF	§5 Results
16	Failure analysis — 3 short cases from <code>regression_diagnoses.md</code>	§5 Results
17	Answers to RQ1 / RQ2 / RQ3 (one bullet each, cite figure number)	§6 Conclusions
18	Recommendations — (cadet-initiated reframing) follow-on directions, master's thesis problem statements, deployment envelope	§6 Conclusions
19	Limitations & future work (single-seed, 25-challenge scope, no LoRA)	§6 Conclusions
20	References (IEEE-formatted) + AI Usage 3-part statement	§7 §8
21	Questions	—

6.2 Q&A prep (Tue, 2 hrs)

Write `qa_prep.md` with rehearsed ≤ 30 -sec answers:

Q	Crisp answer
Why local model not just GPT-4?	Cost, sovereignty (military deploy), reproducibility — and as a hypothesis test about scaffolding's marginal value at sub-frontier capability
Why no LoRA?	Audit showed scaffolding dominates LoRA at this stage; LoRA is the explicit next phase, listed as a master's-thesis problem in §6.2
Why these 25 challenges?	Stratified across web vuln categories from picoCTF + MetaCTF; bounded by free-tier rotation. Document exclusions on slide 9
How does this compare to EnIGMA / Cybench?	EnIGMA uses GPT-4 + Docker exec; we use a 26B local agent + structured lessons. Cybench is a benchmark, not an agent. We're closest to EnIGMA's

Q	Crisp answer
	<i>methodology</i> , with a scaffolding-first variant at smaller scale
Why did KB hurt some challenges?	Three diagnosed mechanisms: context bloat (P3a), stale-lesson cross-talk (v3.4 site-fingerprint), narrow IDOR range default (P5b). See slide 16
Reproducibility?	<code>./scripts/run.sh</code> wraps <code>env</code> (DYLD libomp fix on Apple Silicon), pinned <code>pyproject.toml</code> , public benchmark TSV, raw logs published
Cadet-initiated — why no sponsor?	Personal interest informed by CTF competition experience; mentor (Lt Col Todd) shaped scope. “Recommendations” section is for follow-on research, not operational deployment
What’s the deploy bar?	Slide 18 — categories with solve rate $\geq$ X% are deploy-ready for human-in-loop CTF triage; lower-rate categories need a human-in-loop
Honor / AI use?	Level-5 attributed Generative AI per DF policy; transcript link on slide 20

Practice each answer aloud once.

6.3 Solo run-through (Tue PM, 30 min)

Set a 25-min timer. If you go over, the cut order is: - First cut: one lit-review thread slide (merge two) - Second cut: one methodology slide (collapse 11+12) - **Never cut Results, Recommendations, or Limitations** — those are graded directly.

6.4 Mock with classmate (Wed AM, 30 min)

Have one classmate (Wilson) ask you 5 questions from `qa_prep.md`. Time each answer. Refine.

6.5 Reproducibility README (Wed PM, 1 hr)

`README_HANDOFF.md` at the repo root:

```
# CTF Solver – Reproducibility & Handoff
```

```
**Status:** v3.10 (8 May 2026). Cadet-initiated individual
    capstone.
```

```
**Mentor:** Lt Col Todd, DFCS.
```

```
**Author:** C1C Andrew Kim, andrew.kim@afacademy.af.edu
```

What this is

A web-CTF-solving agent that combines a unified scaffolding layer (structured tool schemas, RAG-augmented prompts, lessons-learned experiential database) with multi-backend LLM support (OpenAI, Anthropic, Ollama, MLX). Released as a basis for follow-on cadet research and as a working tool for picoCTF triage.

Quickstart

```
./scripts/run.sh \  
  --challenge-url <URL> \  
  --challenge-name <NAME> \  
  --description "<one-sentence brief>" \  
  --flag-preset picoctf \  
  --llm-provider ollama --model gemma4:26b \  
  --rag-mode lessons_readonly \  
  --max-steps 15 --history-window-size 16
```

Apple Silicon – required env (segfault prevention)

See ``memory/faiss_libomp_crash.md``. The wrapper sets DYLD_INSERT_LIBRARIES so faiss/torch/sklearn share one libomp.

Run the benchmark

```
./scripts/benchmark.sh  
# Override defaults via env: RUN_OPENAI, RUN_OLLAMA, MAX_STEPS,  
  OPENAI_MODEL.
```

What it solves reliably (May 2026 benchmark)

[Categories-with-rates table from results/metrics.csv]

Known gaps & follow-on work

1. [Category X] – model misidentifies pattern, needs hint Y
2. [Category Y] – observation truncation drops critical signal
3. [Category Z] – schema friction on tool W

Master's-thesis-scope follow-ons

1. RL on top of SFT-LoRA trained on scaffold-aligned trajectories
2. Multi-agent decomposition (recon-agent + exploit-agent + extract-agent)
3. Cross-challenge transfer evaluation for the lessons-DB

Key dependencies

- Python 3.13, faiss-cpu, sentence-transformers, torch
- Ollama (gemma4:26b for local) and/or OpenAI API key
- See pyproject.toml for pinned versions.

Reproducing the benchmark

1. cp bench/challenges.tsv.template bench/challenges.tsv # edit URLs
2. export OPENAI_API_KEY=...
3. ./scripts/benchmark.sh
4. python bench/analyze.py # produces out/figures/*.png

Tests

pytest tests/ # currently 2662 passing

Contact

C1C Andrew Kim – andrew.kim@afacademy.af.edu

Mentor: Lt Col Todd, DFCS

6.6 Submission checklist (Thu PM)

- [] Final paper PDF – all 6 sections present, 5–8 pages
- [] Abstract $\leq$ 250 words
- [] All figures captioned, referenced in text by number
- [] All in-text citations resolve in references list
- [] No "TODO" / "[FILL IN]" / placeholder text (grep)
- [] AI Usage 3-part statement present
- [] PDF rendering verified on a second machine
- [] Slide deck PDF exported
- [] README_HANDOFF.md committed
- [] All run logs preserved under runs/benchmark_*
- [] Repo tagged: git tag v3.10-final
- [] qa_prep.md committed
- [] Team Assessment email drafted (solo cadet – short)

6.7 Submit (Fri 8 May, before 0900)

Subject: [CAPSTONE] DATA 422 Final Report – C1C Andrew Kim

To: justin.merrick@afacademy.af.edu

CC: lt.col.todd@afacademy.af.edu

Sir,

Attached is my final report for DATA 422. Section 8 (AI Usage Statement)

includes the transcript link. The repository tag v3.10-final captures the exact codebase state.

Final brief is on T__ at ____.

V/R,

C1C Andrew Kim

7. Code Changes Required (Summary)

File	New / Modify	Purpose
bench/challenges.tsv	New	Benchmark spec
bench/README.md	New	

File	New / Modify	Purpose
		Scope, exclusions, limitations
scripts/benchmark.sh	New	Hybrid OpenAI + Ollama harness
scripts/extract_metrics.py	New	CSV from run logs
bench/analyze.py	New	5 figures from CSV
refs/synthesis.md	New	Lit-review synthesis paragraphs
methods_inventory.md	New	Novel-methods table by tier
results/regression_diagnoses.md	New	Causal analysis of failed runs
sponsor_notes.md	New	Mentor meeting notes (filename kept for code compat)
final_paper/main.tex + references.bib	New	Paper
presentation/final_brief.pptx	New	Deck
qa_prep.md	New	Q&A defense
README_HANDOFF.md	New	Reproducibility / handoff
Possibly ctf_solver/runner.py	Modify	Add --save-metrics JSON output to make extract_metrics.py simpler

No existing test will break.

8. Rubric Coverage Map

Final Paper Rubric (DS422_EOC_Paper_Rubric.pdf):

A-line	Where you earn it
“Concise summary of problem, methods, and results”	§1 Abstract + §2 Intro
“Synthesizes high-quality research on ‘new topics’”	§3 Lit Review (refs/synthesis.md)
“Effectively integrates coursework with novel methodologies”	§4 Methodology + methods_inventory.md
“Focuses on primary solution; high-quality visuals”	§5 Results (Figures 1–5)
	§6 (with cadet-initiated reframe)

A-line	Where you earn it
“Data-driven conclusions; decision-quality recommendations”	
“All sources cited professionally (IEEE/APA). Conference standard”	§7 + §8

Final Presentation Rubric (Final_Presentation_Rubric.pdf): 1:1 by slide, see §6.1.

Course Letter §6 holistic A: covered by mentor sync (§4.4), polished brief (§6.1–6.4), professional submission (§6.6–6.7), and the codebase artifacts themselves.

9. Risk Register

Risk	Mitigation
OpenAI benchmark fails / rate-limits	Already using gpt-4o-mini (cheap, high quota); fall back to Ollama-only with a note in §5 limitations
FAISS+OpenMP segfault returns	Already patched via DYLD wrapper + memory note
Lt Col Todd unavailable for sync	Send written summary; capture his preferences async via email
Run-time on benchmark exceeds calendar	gpt-4o-mini takes ~2 hr for 75 runs; well within Thu morning
LaTeX template trouble	Fallback to Word IEEE template — content is what’s graded
Brief runs over 25 min	Cut order documented in §6.3
Fresh segfault on a tool I haven’t touched	Check ~/Library/Logs/DiagnosticReports/Python-*.ips, apply DYLD env, log it as known-issue

Appendix A — Quick reference: files you’ll create

```

bench/
├─ README.md
├─ challenges.tsv
├─ analyze.py
scripts/
├─ benchmark.sh
├─ extract_metrics.py
├─ run.sh
refs/
├─ synthesis.md
├─ *.pdf
results/
├─ regression_diagnoses.md

```

(already exists)

(the 7 papers)

```

out/figures/
├─ fig1_solverate.png
├─ fig2_matrix.png
├─ fig3_cdf.png
├─ fig4_validity.png
├─ fig5_stuck.png
final_paper/
├─ main.tex
├─ references.bib
├─ figures/                (symlink → ../out/figures)
presentation/
├─ final_brief.pptx
methods_inventory.md
qa_prep.md
sponsor_notes.md
README_HANDOFF.md

```

Appendix B — Today's 60-minute starter sequence

```

cd "/Users/andrewkim/Documents/School Files/1 Dig Spring Semester/
    DATA422/new_final_project/kim_data_project"
mkdir -p bench refs results out/figures presentation final_paper

# 1. Email Lt Col Todd (2 min)
# Use the template in §4.4

# 2. Pull the 7 papers as PDFs into refs/ (15 min)
# Use Google Scholar; accept the arXiv PDFs

# 3. Build challenges.tsv from your existing lessons docs
ls out/lessons_knowledge/lessons_*.md | sed -E
    's|.*lessons_[0-9]+_|'|' | sed -E 's|_r[0-9]+.*||' | sort
    -u
# Edit bench/challenges.tsv with these names + URLs

# 4. Paste scripts/benchmark.sh + scripts/extract_metrics.py from
    §4.2
chmod +x scripts/benchmark.sh

# 5. Dry-run on one challenge with OpenAI gpt-4o-mini
RUN_OLLAMA=0 OPENAI_MODEL=gpt-4o-mini MAX_STEPS=10 ./scripts/
    benchmark.sh
cat runs/benchmark_*/metrics.csv

# 6. If clean: full sweep before bed
./scripts/benchmark.sh

```

You're set up to wake up Thursday with the full benchmark CSV in hand.